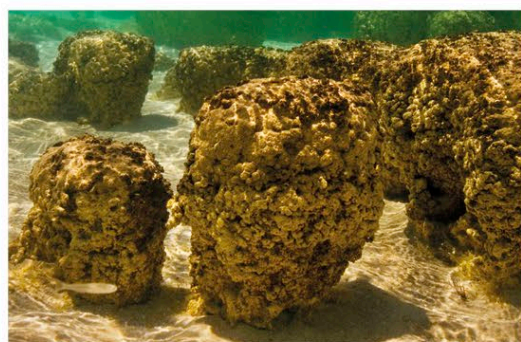


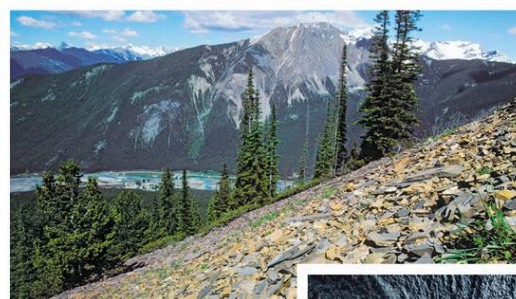


< STROMATOLITES

For billions of years these layered structures have built up in shallow tropical seas by alternating layers of sediment and sheets of micro-organisms, including cyanobacteria (blue-green algae).

Around 650 million years ago, fossils of various frond- and disc-shaped, soft-bodied, and mostly sessile organisms appeared. These marine animals were known as the Ediacaran Fauna. By the beginning of the Cambrian Period, 545 million years ago, numerous multi-celled marine organisms had evolved, including burrowing worms and a variety of small, shelled molluscs, whose bodies had muscle tissues and organs such as gills for respiration. Around 510 million years ago, the first vertebrates appeared, with an internal skeletal support for the body. By late Devonian times, around 380 million years ago, vertebrates had begun to emerge from the seas onto land.

From the first appearance of simple life, it took at least 2.7 billion years before complex life forms appeared. The fossil of a microscopic and multi-celled red algae called *Bangiomorpha* provides the first evidence of the existence of specialized cells. These cells evolved for sexual reproduction, and also for the development of a holdfast to attach the algae to the sea floor.



^ BURGESS SHALE

The fossils of the Burgess Shale in Canada show that marine life diversified rapidly in Cambrian times, from sponges and arthropods to vertebrates.

> WIWAXIA

Found at the Burgess Shale, *Wiwaxia* was a 5 cm (2 in) long, mollusc-like seabed crawler with spines and scales, and a soft undersurface.

